Valluri Keerthi Ram

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Education

Amrita Vishwa Vidyapeetham — ASEB, Bangalore
BTech, Computer Science and Engineering, CGPA: 8.74 · AEEE rank: 934 — 2023-Present

Central Board of Secondary Education
Class 12, Percentage: 72.6% — 2021-2023
Class 10, Percentage: 84% — 2021

Experience

AlgoUniversity — June 2026 – July 2026
Software Engineering Intern (Human-Centered AI & LLM Systems) – Agentic Runtimes

- Architected an AST-validated query runtime and human-in-the-loop reasoning engine over relational databases (PostgreSQL, MySQL), exposing high-throughput FastAPI REST microservices.
- Constructed an automated model evaluation harness and synthetic benchmark suite with closed-loop self-repair, diagnosing failure modes and elevating agent task execution fidelity from 72% to 94%.
- Implemented zero-trust schema invariants, structured tool calling, and guardrails, preventing prompt injection, data drift, and hallucinations across production user-facing interfaces.

Projects

Scriptio Continua: Multimodal Tokenization & Accessible NLP (IITCEE 2027) — Jan 2026 – Present
Python • Foundation Models • vLLM • PyTorch • Linear-Chain CRF • Cognitive Accessibility

- Investigated tokenization failure modes and quadratic compute scaling in open-weight LLMs (Gemma, Qwen, DeepSeek R1 32B), engineering a hybrid 1D-CNN + BiLSTM + CRF model achieving 0.9622 BERTScore F1.
- Benchmarked directly against open-weight LLMs and vLLM inference runtimes, demonstrating competitive segmentation accuracy with an 85% compute overhead reduction for continuous language streams (IITCEE 2027).

Emotion & Sentiment Modeling in Low-Resource Indic Languages (ICKECS 2026) — Aug 2025 – Nov 2025
Python • Multilingual Transformers • IndicBERT • XLM-RoBERTa • Human-Centered AI • NLP

- Fine-tuned multilingual transformer models (XLM-RoBERTa, IndicBERT) to concurrently perform multi-task emotion classification and semantic sentiment analysis over low-resource Dravidian text.
- Mitigated severe corpus class imbalance through contextual augmentation pipelines, analyzing cross-lingual representation alignment and achieving a macro F1 of 0.784 (IEEE ICKECS 2026).

Cross-Modality 3D Deep Generative Synthesis & Clinical AI (FTNCT 2026) — July 2026 – Present
Python • Medical Imaging • Computer Vision • Generative AI • PyTorch • Clinician-in-the-Loop

- Architected a resource-efficient 3D deep generative synthesis pipeline for cross-modality volumetric translation (CT-to-MRI), optimizing latent representations and spatial consistency to resolve imaging bottlenecks.
- Engineered automated ingestion and normalization pipelines with rigorous data validation assertions, accelerating evaluation workflows across clinical imaging records (FTNCT 2026).

Research & Publications (First Authors)

- **IITCEE 2027:** Breaking the Continuum: Automatic Word Segmentation for Scriptio Continua Languages (Under Publication)
- **FTNCT 2026:** Deep Computer Vision Approaches for Brain CT-to-MR Image Synthesis in Cross-Modality Medical Imaging (Under Publication)
- **ICKECS 2026 (IEEE):** Emotion and Sentiment Detection in Telugu Text Using Multilingual Transformer Models
- **CICCS 2025:** Enhancing Cyber Threat Intelligence Using Latent Dirichlet Allocation (LDA)
- **CINE 2026 (IEEE):** Prediction of Groundwater Level for Semi-Arid North-Western India Using Rainfall Data: A Comparative Analysis
- **ICCTWC 2026:** Crypto Trading Platform Using Probability for Scalping and Compounding
- **Journal of Engineering Design and Analysis (2026):** Elevator Optimization With Real-World Constraint: Weight-Aware Algorithm

Certifications

AWS Certified Cloud Practitioner — May 2026
Credentials: 1118bc1d19af4214a1d5da0f72b73039 — AWS

Introduction to Modern AI — Cisco

Project Management Foundations — Oct 2024
Credentials: 6c22df1ec90364652f3ff5e3aaacbf889196fac08c5d03a9b495b575aa3f54f4 — PMI & NASBA

Skills & Interests

- **Human-Centered AI & NLP:** Multilingual Transformers, LLMs, Tokenization, Low-Resource NLP, vLLM
- **Languages & Frameworks:** Python, PyTorch, Hugging Face, FastAPI, SQL (PostgreSQL), REST APIs
- **Tools, Cloud & Evaluation:** AWS Certified, Docker, Git, Model Evaluation, Benchmarking, Linux